

KOGALYM, Russian Federation

WMO: 237480

Lat: **62.1975N**Lon: **74.5388E**Elev: **68**StdP: **100.52**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.1	-39.1	-44.2	0.0	-41.6	-41.4	0.1	-38.8	9.3	-7.9	8.7	-9.4	1.9	20	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	29.0	19.8	26.9	19.1	24.8	18.0	21.0	26.8	20.0	25.0	18.8	23.2	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	23.2	18.0	13.1	22.4	17.0	12.2	21.9	61.3	26.7	57.6	25.2	53.8	22.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.4	DB	-44.4	30.5	4.3	1.8	-47.5	31.8	-50.1	32.9	-52.5	33.9	-55.7	35.3	(4)
(5)			WB	-44.0	22.5	3.8	1.4	-46.8	23.4	-49.0	24.2	-51.1	25.0	-53.8	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-22.0	-20.4	-9.8	-4.0	4.1	14.8	18.3	13.6	7.6	-1.4	-12.4	-17.8
	DBStd	15.32	10.03	9.24	7.24	6.71	5.90	5.41	4.13	3.53	4.28	5.44	8.76	9.55
	HDD10.0	5096	992	852	613	421	204	19	1	7	98	353	673	863
	HDD18.3	7633	1250	1086	872	671	443	129	53	151	323	611	923	1121
	CDD10.0	589	0	0	0	0	21	164	258	119	26	1	0	0
	CDD18.3	84	0	0	0	0	2	24	53	5	1	0	0	0
	CDH23.3	734	0	0	0	0	21	212	465	35	2	0	0	0
	CDH26.7	186	0	0	0	0	4	49	129	4	0	0	0	0
Wind	WSAvg	3.8	3.3	3.4	4.1	4.2	4.3	3.9	3.6	3.7	3.8	4.1	3.6	3.5
Precipitation	PrecAvg	824	53	43	59	54	64	78	71	89	74	86	85	67
	PrecMax	1031	102	83	151	85	150	137	170	151	146	150	138	135
	PrecMin	652	15	11	3	2	33	35	30	34	29	37	31	27
	PrecStd	97	22	17	28	20	27	23	29	28	27	29	27	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	1.2	4.8	11.2	26.0	30.6	32.0	26.5	21.8	12.9	3.3	0.0
		MCWB	-0.3	0.1	1.5	7.2	17.0	18.6	20.9	20.0	15.5	9.2	3.0	-0.4
	2%	DB	-3.2	-2.2	2.4	7.4	21.1	28.0	29.9	24.0	18.2	9.2	1.5	-1.2
		MCWB	-3.6	-2.9	0.4	4.4	14.1	19.0	20.0	17.8	14.0	7.3	1.2	-1.5
	5%	DB	-5.8	-5.5	0.8	5.8	17.1	25.9	28.0	22.0	16.0	7.4	0.6	-3.2
		MCWB	-6.1	-6.2	-0.7	3.5	11.8	18.1	20.0	16.8	12.7	6.3	0.3	-3.6
	10%	DB	-8.2	-7.6	-0.4	4.2	13.2	23.7	26.0	20.1	14.0	5.5	-0.8	-5.4
		MCWB	-8.5	-8.2	-1.6	2.3	9.4	17.2	18.8	15.6	11.7	4.7	-1.2	-5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.2	1.9	7.7	17.6	22.2	22.5	21.3	15.8	9.5	3.0	-0.6
		MCDB	-0.7	1.3	4.2	10.6	24.7	26.7	28.4	25.4	19.1	12.1	3.3	-0.4
	2%	WB	-3.7	-2.9	0.5	5.1	14.7	20.3	21.3	19.1	14.4	7.6	1.2	-1.7
		MCDB	-3.3	-2.2	1.9	6.7	21.1	25.1	27.4	22.7	17.5	8.8	1.3	-1.5
	5%	WB	-6.1	-6.1	-0.4	3.6	12.3	19.1	20.6	17.6	13.2	6.3	0.2	-3.6
		MCDB	-5.9	-5.5	0.8	5.3	16.5	23.2	26.4	20.7	15.6	7.3	0.4	-3.3
	10%	WB	-8.6	-8.2	-1.6	2.5	9.7	18.0	19.6	16.3	11.9	4.6	-1.1	-5.8
		MCDB	-8.3	-7.6	-0.6	4.1	13.3	22.2	24.8	19.3	13.9	5.3	-0.9	-5.5
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.0	9.4	9.1	8.4	8.9	9.1	7.9	6.4	5.0	6.8	7.5
		MCDBR	8.0	7.6	8.1	9.0	13.4	11.9	12.1	11.0	9.2	5.9	5.5	6.1
		MCWBR	8.0	7.4	6.9	6.8	7.9	5.4	5.1	5.9	5.6	4.3	5.4	6.1
	5% WB	MCDBR	8.3	7.7	8.7	7.9	12.7	10.2	10.6	9.7	8.8	5.5	5.4	6.3
		MCWBR	8.3	7.5	7.9	6.2	7.9	5.7	5.0	5.9	5.7	4.2	5.5	6.3
Clear-Sky Solar Irradiance	taub	0.246	0.291	0.344	0.365	0.378	0.394	0.422	0.409	0.354	0.320	0.249	0.233	
	taud	1.939	2.016	1.966	2.053	2.296	2.365	2.274	2.279	2.400	2.240	1.990	1.892	
	Ebn at Noon	387	630	728	807	838	830	789	763	747	607	378	208	
	Edh at Noon	39	76	117	132	114	109	116	107	78	62	36	23	
All-Sky Solar Radiation	RadAvg	0.31	1.17	2.80	4.64	5.26	5.36	5.07	3.51	2.13	0.87	0.29	0.14	
	RadStd	0.02	0.10	0.16	0.43	0.37	0.44	0.55	0.41	0.25	0.12	0.07	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-368	N/A	N/A

Nomenclature: See separate page